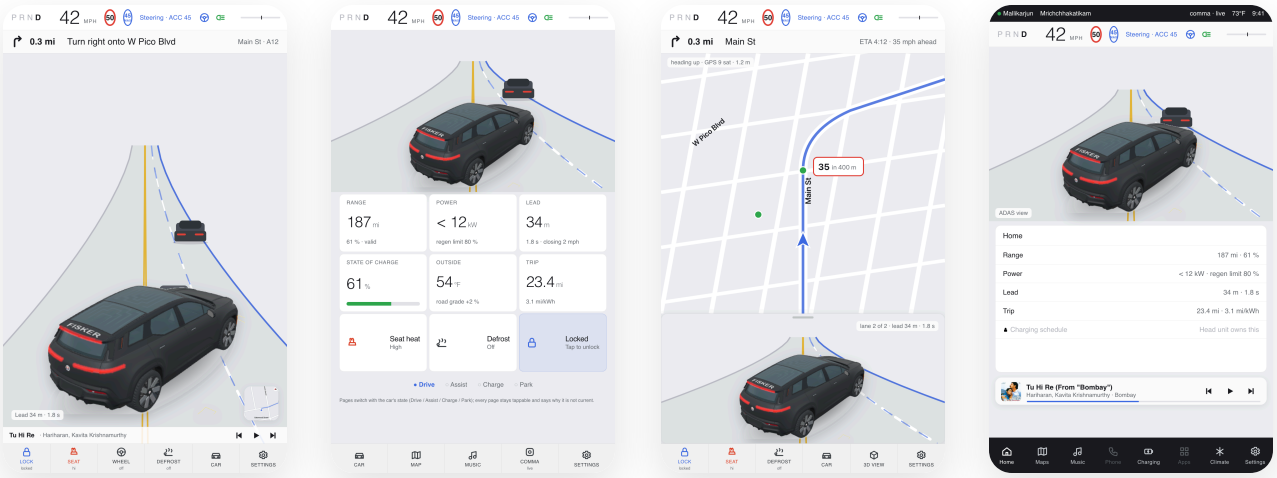


Pulse HUD UX options

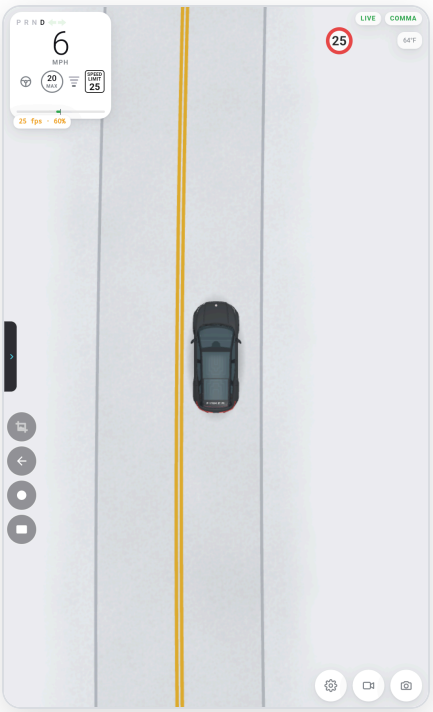
Four portrait layouts make the Pulse HUD glanceable: the current experience of the Mrichchhakatikam HUD on the Fisker Ocean's Pulse screen scored against automotive display guidance, four layouts that expose the same functions, and the enhancements the car, the comma and the map already make possible.

Research report with mockups · 1 November 2026 · Sources: the repository on branch `pulse-coexist` (working tree of 31 October 2026), NHTSA and Euro NCAP guidelines, manufacturer manuals, published HMI research, owner communities. Mockups are drawn at Pulse's measured 1080 × 1784 portrait and 1784 × 1080 landscape.

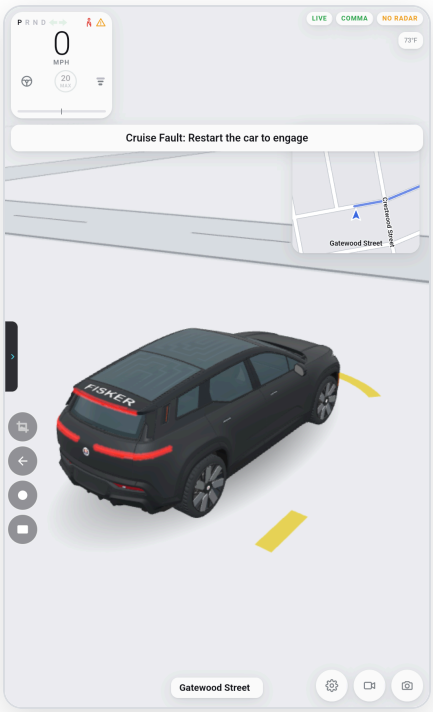
The Mrichchhakatikam HUD today is a full-screen three.js world with about fifteen overlays pinned to the screen edges, no home screen, three 54 px buttons and an invisible tap target on the car as its only persistent controls, and **nothing in the page locked by speed** (the parking auto-view is the one speed-driven change; the moving-car refusals all live on the comma and in the Android app). Automotive guidance converges on a small set of numbers the page can be scored against: **no single glance over 2.0 s and no more than 12 s of eyes-off-road per task** (NHTSA), **speed, speed limit, ADAS state and exterior-light state inside a 30-degree cone**, **touch targets of at least 10 × 10 mm with 4 mm gaps**, **at most two menu steps from the driving screen**, and a per se lock-out of video, free text entry and self-scrolling text whenever the car is not in Park. The stock Ocean drives in portrait “Control Mode” with a fixed status bar at the top and a fixed main menu at the bottom; landscape “Hollywood Mode” exists only in Park and the car refuses to leave Park in it, so the orientation the task brief assumed is backwards for the stock car and remains the first thing to verify on Pulse. Of the four layouts proposed here (Glance Strip, Tile Dash, Map Centre, Ocean Native), the Glance Strip is the right base: it is the 96 px status strip DESIGN.md already specifies but never built, plus the fixed bottom bar Euro NCAP calls “direct touch input”, and it keeps the 3D world as the centre while shrinking everything else to one line while moving. The Tile Dash and Map Centre earn their place as the parked and charging pages and as the landscape (Hollywood) variant, and the Ocean Native layout is best used as a skin for the status bar and menu order rather than as a structure. The cheapest, highest-value change is not a layout at all: a page-side “not in Park” policy using `VCU_GearSig` (already received on IBUS1) that greys the camera window, the Comma tab’s command box, the log tail and the parked-only rows, mirroring the comma’s own refusals. Beyond that, the car already delivers about **536 decoded signals nobody draws**, including remaining range, trip and odometer, the car’s own driver-monitoring state, the red-light warning and green reminder, a camera-blinded flag and GPS fix quality, all page-only work; a `gen_ibus.py` re-run would add the 12 V battery, TPMS lamps, heated-wheel and auto-hold read-back, steering-wheel buttons and the screen rotator; and radar stays a hardware project.

Repository citations below point at files relative to the repository root (`../` from `reports/`). Line numbers refer to the working tree on `pulse-coexist` as of 2026-10-31.

Today on the car



Driving at 6 mph: the floating status card, speed-limit chip, lanes from the map road, the four-button view strip and the three-button dock. Screenshot of the app on Pulse, 2026-10-31.



Parked with a cruise-fault alert: the alert band under the status card, the new 2D mini-map at top right, the road name, the parking arcs. Same session.

Both screenshots were taken with `adb screencap` from the Pulse board while the app was running in the car on 2026-10-31. They are **1080 × 1784 portrait**, which settles the first row of the open-questions table for the Android surface in Control Mode: Pulse renders portrait at 1080 × 1784, and the mockups in this document are drawn at that size. Whether the surface rotates in Hollywood Mode is still unverified. The mockups keep the app's light theme, type and card language (DESIGN.md), reuse the car renders from these screenshots, and draw the road scene schematically; the numbers on them (range, lead distance, charge rate) are illustrative.

Today's page is a world view with fifteen overlays and no speed lock

The root of the app is a full-screen canvas with `touch-action: none` on which the ego Ocean, lanes, the map road or the camera road, fused objects, lights, signs and parking arcs are drawn, and every other surface is an overlay reached by one gesture from it: the gear button opens a seven-tab settings sheet, a tap on the car opens car controls, the `comma` chip opens the Comma tab, the power bar opens the energy trace and then the Energy dialog, the video button opens a draggable camera window, and Android Back opens a native menu (`static/index.html#L30–L137`; `static/js/main.js#L488–L515`; `static/js/settings.js#L28–L53`). The status card sits top-left at 300 px wide in landscape and shrinks to a **172 px floating square** in portrait; the alerts band hangs under the next-turn card at top-centre; chips, road-info rows and the new 2D mini-map stack top-right; the music card sits bottom-left, the road name bottom-centre and the three-button view dock bottom-right (`static/app.css#L94–L106`; `static/app.css#L355–L379`; `static/js/minimap.js#L1–L9`). The top-right corner is the crowded one: chips at 16 px, road-info rows growing downward from 52 px, and the map card at `max(130px, --`

`drive-h) + 12px`, so a traffic light, two park-distance rows, a hazard row and the outside temperature can push the road-info stack past the map card's top; nothing in the repo records whether that overlap occurs on the car (`static/app.css#L180-L184`; `static/app.css#L562-L572`).

Everything driven by the ADAS mirror on IBUS2 and the body messages on IBUS1 works from the car alone: speed, gear, telltales, lanes, objects, signs, lights, parking sensors, the map road and the mini-map (which needs the TBOX fix), and the car-state read-outs (`INTERNALS.md#L171-L183`; `android/README.md#L4-L7`). The comma adds replay, radar tracks, openpilot alerts and MADS text, the camera window and the five ACC settings, and the Android bridge (`window.WebHudApp`) adds music and navigation cards, car-control sends, comma management, self-update and device location; the code is required to keep working with neither (`static/js/settings.js#L10-L14`; `static/js/infotainment.js#L1-L6`; `static/js/cancmd.js#L73-L77`; `CLAUDE.md`). The Android README states the design intent for absent devices plainly: "Nothing on screen waits for it or says it's missing: no connecting card, no reconnecting pill, just a comma chip while it's there" (`android/README.md#L10-L12`).

Car controls are the one place with a designed structure. A tap on the car slides up a dock of ten car-part categories and five system ones, the camera goes top-down with the roof faded, and a sheet opens on the right in landscape or as a bottom sheet with **three detents of 448, 848 and 1360 frame pixels** in portrait; each control is tagged Sends, Reads only, Via comma, Not available or Demo, and an unavailable control is greyed with a lock, its reason in a toast (`static/js/carcontrols.js#L27-L40`; `static/app.css#L420-L427`; `DESIGN.md#L42-L43`). Of the ten part categories only Doors & Windows, Climate (minus temperatures), Seats (heat, memory, easy entry) and Lighting (headlights, ADB) send anything; Driving, Energy and Display are read-only with every setter greyed as head-unit-owned, Assist is read-only except the five comma-routed ACC settings, and Audio and Service are placeholders (`static/js/carcatalog.js#L99-L454`). The settings sheet, by contrast, is a developer's instrument panel: of roughly 45 Display switches about a third are calibration or demo toggles, three tabs (CAN settings, Driving, Playback) show only a banner without the comma, and the Comma tab is SSH administration with a free-text command box (`static/js/settings.js#L435-L473`; `static/js/comma.js#L1-L17`).

The moving-car guards are real but they are all outside the page. The comma refuses CAN overrides, SSH key changes and replay while moving; the Android app refuses comma jobs and self-install above walking pace; the Assist tab's ACC settings grey with "Parked only" (`README.md#L115-L116`; `android/README.md#L170-L172`; `static/js/accsettings.js#L32`). The page itself sends window, sunroof, liftgate, lock and seat-heat requests at any speed, shows the comma camera window at any speed, keeps the Comma tab's command box and auto-tailing log reachable at any speed, and lets the settings sheet and car mode stack on top of each other, because nothing closes one when the other opens (`static/js/cancmd.js#L67-L77`; `static/js/main.js#L435-L459`). The one designed moving-versus-parked distinction is DESIGN.md's principle that cells are for what you do while driving and rows are for settings reached when parked, applied by control shape rather than by a lock (`DESIGN.md#L28-L46`). The authors' own open list is about performance and correctness, not layout: the car-mode status strip "is not built (the HUD's status card stays)", the on-car cards in landscape still use the old layout, the head unit's budget with an 815k-triangle model is unchecked, Pulse drops about half of IBUS1 and two thirds of IBUS2 frames, and no screenshot or measured layout of the real screen exists in the repo (`DESIGN.md#L237-L239`; `CAR_CONTROLS_PLAN.md#L137-L144`; `android/README.md#L151-L154`).

Guidance converges on 2 s glances, a 30-degree cone, 10 mm targets and two menu steps

The numbers that recur across the guidelines are few and testable. NHTSA's 2013 Phase 1 guidelines accept a task for use while driving only if, for 21 of 24 participants, no more than 15 % of glances exceed 2.0 s, the mean

glance is at or below 2.0 s, and total off-road glance time is at or below **12.0 s**; “driving” is defined as the transmission not being in Park, and NHTSA explicitly kept Park as the line despite requests for a “stopped but not in Park” exception (NHTSA Guidelines, 78 FR 24818). The same document lists the per se lock-outs: functions not intended for use while driving, manual text entry, video (with an exception for the rear view in reverse that must end at 10 mph, 10 m or 10 s), non-driving images (maps and driving-related images are exempt), automatically scrolling text, and text to be read from books, web pages or messages; it adds that photorealistic, satellite or **three-dimensional map detail “is not recommended”**, that device feedback should arrive within 0.25 s with a “working” indication if a response takes over 2.0 s, that only one hand may be needed, and that dynamic non-safety information must be dimmable or blankable while driving (NHTSA Guidelines, secs. V.F, V.K, V.I, V.L). SAE J2364’s older rule is that a navigation task taking more than 15 s when measured parked should not be available in motion, with the author arguing for about 10 s (Green, 15-Second Rule, TRID).

Euro NCAP’s 2026 Driver Engagement protocol turns these into layout rules. Essential driving information (“vehicle speed, applicable speed limit, ADAS status, and all exterior light status”) must be in the direct line of sight, within ±30 degrees laterally and above a plane 30 degrees below the ocular reference point; touch targets must be **at least 10 × 10 mm with 4 mm separation**; “direct touch input” is defined by example as “a fixed bar with digital controls at the bottom of an in-vehicle infotainment screen”; menu-based input is capped at **2 steps counted from the home screen**, where the home screen is “the screen used while driving”; and a long distraction is a single 3-4 s gaze away with cumulative 10 s in 30 s counting as short distraction (Euro NCAP Driver Engagement v1.2). The same protocol requires that a non-functional driver-monitoring system be announced visually within 10 s with the notice persisting or repeating every 10 minutes, which matters because the Ocean’s own driver camera state is already decoded and unread (see the enhancements section) (Euro NCAP Driver Engagement v1.2, sec. 1.1.1.2). For legibility, ISO 15008 sets character height as an angle; the readable secondary literature puts the moving-vehicle recommendation at **24 arcmin** with a 14-15 arcmin floor, and the standard’s clause values themselves are paywalled and recorded as a gap (Assessment of Automotive Visual Display Guidelines; ISO 15008:2017 preview).

Production cars apply these in two patterns. Tesla, Rivian and Lucid split a landscape screen into a driver-side car-status visualization, a passenger-side map and a fixed bottom bar, with Tesla letting the owner drag the boundary and the Cybertruck offering a one-third or full-screen visualization (Tesla Model Y manual; InsideEVs Rivian review; Tesla Oracle Cybertruck). Polestar, Volvo, Mercedes and Hyundai use a tile home with a persistent climate bar at the bottom and status at the top, Mercedes keeping the map always visible under suggested tiles (Polestar 2 manual; carsales on Zero Layer). The loudest criticisms are of forcing speed and basic status onto the centre screen (the Volvo EX30, whose driver camera then warns the driver for looking at it), of screen-only vents and multi-tap fan speed, and of surrounding-traffic visualisations that are “useful during assisted highway driving but busy around town” (InsideEVs EX30 roundup; InsideEVs Rivian review). Polestar closes its camera view “automatically when the car is driving at a certain speed”, Tesla disabled games in motion after NHTSA opened a probe, and Mercedes uses the driver camera to block moving pictures on the passenger display (Polestar 2 manual; Engadget; MercedesBlog).

The research on 3D world views says what is shown matters more than how it is rendered. Colley et al. found “almost no significant differences” between a 2D rendering and the same scene on lightfield and holographic displays, and in a 216-participant study found that visualising other agents’ predicted paths was rated worse and raised workload (NASA-TLX 12.40 vs 9.89, $p = 0.001$) while showing detections raised neither trust nor understanding significantly; STADT:up’s 2026 Berlin and VR studies found trust rose when the vehicle showed what it perceived, that more simultaneous interfaces raised cognitive load, and recommended showing information when needed and holding it back when disruptive (Colley et al. 2023; Colley et al. 2022; Ergosign STADT:up). The openpilot forks are a ranked wish list of what driving-assist users add when they can: mode-coloured path edges, blind-spot cues, lead distance and time under the chevron, a speed-limit display with

warning, the road name, a standstill timer, green-light and lead-departure chimes, hide-speed, and a standby mode that blanks the screen onroad and wakes for alerts (sunnypilot Visuals; FrogPilot wiki). The S3XY Dash for Teslas, the best-documented aftermarket companion screen, is a grid of 1×1, 1×2 and 2×1 stat widgets with pages that switch automatically on Drive, Autopilot or charging, one-tap actions whose state shows on the tile, and a light theme added for windshield glare (enhauto S3XY Dash; enhauto tutorial).

Scoring the current page against those numbers

Converting the guideline millimetres to pixels needs the panel's size and resolution, and the repo states neither outright; DESIGN.md designs for a **1080 × 1920 frame** and the Ocean's screen is **17.1 inches**, which gives about 129 px per inch if the frame fills the panel, so **10 mm is about 51 px and 4 mm about 20 px** (`DESIGN.md#L3–L5`; GearJunkie). On that basis the car-controls language passes comfortably (minimum hit 72 frame px is about 14 mm, dock items 80 × 96, rows 80, slider thumb 88 × 52) (`DESIGN.md#L102–L113`), the view dock's 54 px buttons pass by half a millimetre, and the **music card's 44 px transport buttons fail at about 8.7 mm** (`static/app.css#L189–L199`; `static/app.css#L315–L333`). The 64 × 36 switch passes only if the whole 80 px row is the target. The 3D car's tap zones have no defined size at all; whether a lamp zone is 10 mm wide depends on the camera distance. The speed numeral is capped at 118 px in landscape and lives in a 172 px square in portrait, far above the 24 arcmin moving minimum at any plausible reach; the alert text sizes are not recorded (`static/app.css#L121–L123`).

Menu depth is mostly within the two-step cap when counted from the HUD: climate is tap-the-car then Climate, locks are tap-the-car then Doors, energy details are power bar then Details; but the first step is an unlabelled, undiscoverable target, and the settings sheet's tab-plus-section structure makes theme, map download or a comma restart three steps (`static/js/main.js#L503–L511`; `static/js/settings.js#L299–L331`). Response time is handled by design (event sends go out as three frames 20 ms apart, the Pending state fades the glyph for up to 1.5 s), though the 1.5 s pending fade is specified in DESIGN.md and has no `pending` class in the code (`static/js/cancmd.js#L4–L18`; `DESIGN.md#L137–L148`). Against the lock-out list the page fails on four counts: the comma camera window is video shown at any speed (the forward road camera is driving-related but no exception covers it, and Polestar's precedent is to close it above a speed), the Comma tab's Debug command and the comma-address dialog are manual text entry, the log tail is self-scrolling text, and the Signals and Comma tabs are long reading; the comma's refusals stop the action but not the reading or typing (`static/js/roadcam.js#L1–L7`; `static/js/comma.js#L569–L579`; `android/app/src/main/java/ai/sunnypilot/webhud/MainActivity.kt#L442–L486`). The 3D scene itself is a driving-related image and therefore exempt, though NHTSA's "not recommended" for 3D map detail argues for keeping the scene abstract, which the research above also supports.

Orientation is the open question: Control Mode drives in portrait, Hollywood parks in landscape

The notes disagree on which way the screen faces while driving, and the disagreement decides every layout. The task brief assumed landscape; the repo's design document specifies car controls "on the Fisker Ocean's portrait center screen, 1080 × 1920", the CSS carries both orientations, and CAR_CONTROLS_PLAN only says that on a short landscape screen the car lies nose-right (`DESIGN.md#L3–L5`; `CAR_CONTROLS_PLAN.md#L45–L46`). The stock car's manual settles it for the stock software: the main menu runs "along the left side of the Central Touchscreen (or the bottom when in Control Mode)", Control Mode's regions are the status bar, "ADAS and camera views", the main menu, maps, an active window and media/phone, and the rotation rules are that "the Central Touchscreen

can only be rotated when the vehicle is in P (Park)" and "the Ocean can only be shifted out of P (Park) when the Central Touchscreen is in Control mode" (Fisker manual, Central Touchscreen Overview; Control and Hollywood Modes). Reviewers describe Control Mode as the portrait driving view and Hollywood as 16:9 for parked video; the rotating screen is only on Ocean One and Extreme, and lower trims have a fixed display (GearJunkie; CompleteCar). Since the bankruptcy, Hollywood Mode is listed by the FOA as not working and is sold back by OV Loop as a paid feature in release 2.6, which makes landscape rarer still on real cars (AUTO Connected Car News; OV Loop 2.6 release notes).

Pulse complicates rather than resolves this. It injects its picture over the ICC's display link (the install taps the HSD cables and pin 28 of the ICC connector) and inherits the Rivian EV Play model of whole-screen switching between the native UI and Android with touch passed through (oceanlinkpulse.com; InsideEVs EV Play). Whether Pulse's Android framebuffer rotates when the panel does, which orientation it reports, and whether the ICC's Park interlock still governs the panel when Pulse owns the picture are all undocumented; the only hints are upstream's note that the camera window saves its geometry "as fractions of the room, so they persist across reloads and rotations" and the Android README's "rotation keeps the page", which show that upstream expected the page to be rotated on the car (moistpope PR #4; `android/README.md#L53–L57`). The bus even carries a screen-rotator message, `Rotator_0x587`, but it is not in the receive list (`dbc/fisker_ocean_ibus.dbc`). The practical conclusion is to design **portrait as the driving layout and landscape as a parked variant**, which is also what the stock interlock enforces, and to treat each option's landscape sketch as a Hollywood-style parked page. The first session on the car should record `wm size` and `wm density` from Pulse's Android in Control Mode, whether the panel and the WebView rotate, the measured viewing angle from the driver's eye point to the top and bottom of the panel (to find which band of the portrait screen is inside Euro NCAP's 30-degree cone), and whether `Rotator_0x587` is broadcast.

Four layouts expose the same functions in different ways

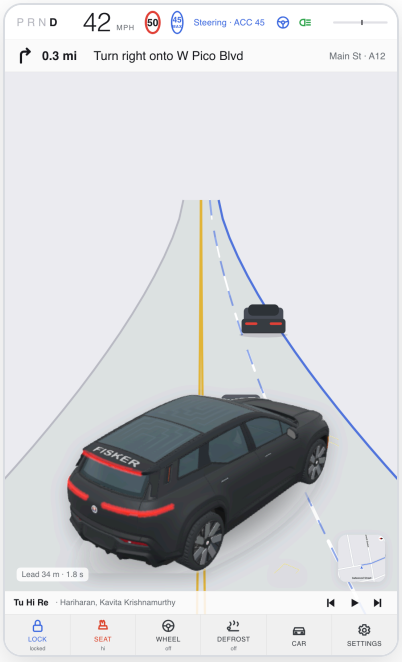
All four options share one policy layer that none of them can do without: a page-side definition of "driving" and a lock-out that greys rather than hides. The car reports gear on IBUS1 (`VCU_GearSig` in `VCU_0x234`, 10 ms) and speed (`ESP_VehSpd`, 20 ms), both already decoded and shown (`research_notes/Pulse%20HUD%20UX%20options/available_data.md`; `static/js/hud.js#L124–L134`). NHTSA's line is Park, so the page should define `driving = gear !== P`, fall back to `speed > 0` when gear is unknown but data is live, and treat the offline state (no driving data at all, the plain-browser mockup) as parked so that the design page and the browser mockup keep working without the bridge or the comma (NHTSA Guidelines, Definitions; `static/js/main.js#L317–L335`). While driving, the page would close the comma camera window (Polestar's precedent), grey the Comma, Signals, CAN settings and Playback tabs, the Debug command box, the address dialog and every `rows`-shaped parked setting, with the reason "Parked only" in the same lock-and-toast language the catalog already uses, and leave the `cells` (climate, seat heat, lock, windows) usable (`DESIGN.md#L42–L46`; `static/js/carcontrols.js#L805–L822`). Whether the BCM itself refuses window, sunroof or liftgate requests in motion is not recorded and should be tested before deciding whether those cells also lock (`research_notes/Pulse%20HUD%20UX%20options/current_app_ux.md`). No option below adds a CAN send: every quick action uses signals inside the seven allow-listed IBUS1 messages (`0x4E` lamps and locks, `0x530`/`0x534` climate, `0x528` seat and wheel heat, `0x52` liftgate, `0x90` lock modes, `0x533` seats), and the head unit's cyclic state messages and the audio frames stay untouched (`tools/gen_ibus.py#L67–L77`; `CLAUDE.md`).

A second shared element is a **fixed status strip** carrying exactly Euro NCAP's essential set: speed, speed limit, ADAS state, exterior-light state, plus gear and the red/amber telltales. DESIGN.md already reserves 96 frame pixels for it at the top of the 1080 × 1920 frame and lists it as unbuilt (`DESIGN.md#L53–L66`; `DESIGN.md#L237–L239`).

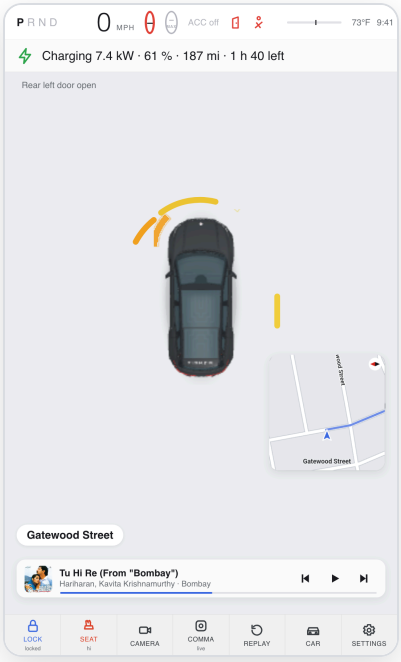
Placing it at the top matches the stock Ocean's fixed status bar and is the band most likely to be inside the 30-degree cone on a tall portrait panel; it is DOM, not canvas, so it costs nothing per frame, and it must not use backdrop blur (the perf meter already strips blur first under load) (`static/app.css#L57–L59`; `static/js/perf.js#L9–L17`). Because the Ocean has its own cluster with a speedometer and a Fi-Pilot indicator, the strip mirrors rather than replaces, which avoids the EX30's failure mode of making the driver look down for speed (Fisker manual, Driver Display).

Option A: Glance Strip (glance-first driving layout)

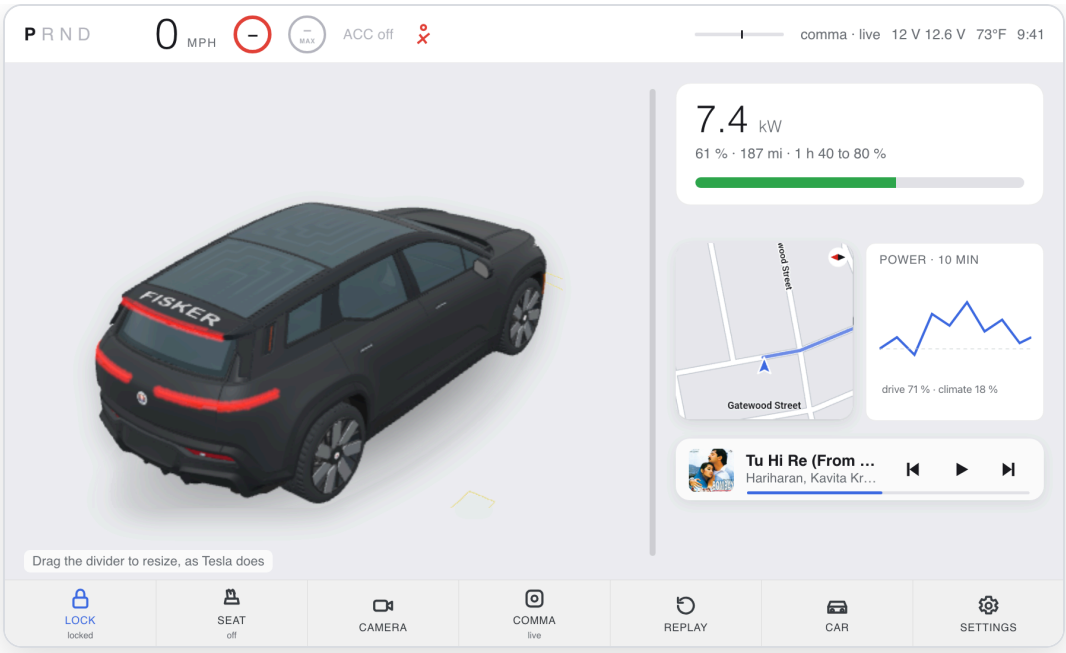
The Glance Strip keeps the 3D world as the whole middle of the screen and reduces everything else to a strip, a band and a bar. While driving, the strip at the top holds speed, limit, ACC set speed and gap bars, the MADS/ADAS icon, beam and light state, gear and telltales; the band below it shows the next turn or the current alert (one at a time, openpilot's three-tier ladder mapped to size and colour); the music card collapses to one 64 px line; the 2D map shrinks to a 120 px square pill with the road name beside it; and a fixed bottom bar of six 160 × 112 frame-pixel targets carries lock, driver seat heat, steering-wheel heat, max defrost, Car (opens car mode) and the gear. Parked (gear P), the bar gains the camera, comma and replay buttons, the music card returns to its full grid, the map pill expands to the current 320 px card, and the car controls open exactly as today.



Driving (gear not P): strip, one-line band, 3D world, collapsed music, map pill, six-button bar



Parked (gear P): charge line, full music card, big map, camera / comma / replay on the bar

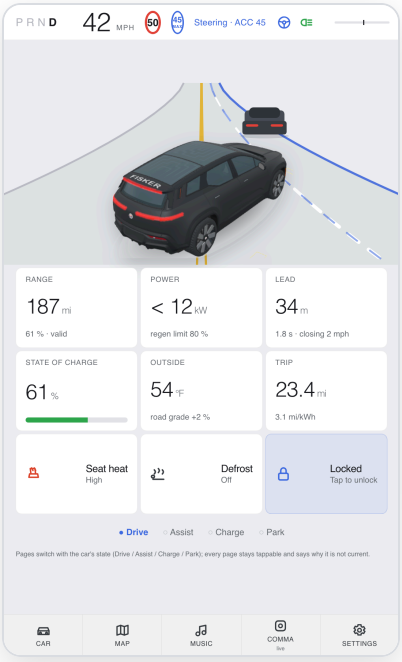


Landscape, parked only (Hollywood variant): 3D car with a draggable divider, charge, map, power, music

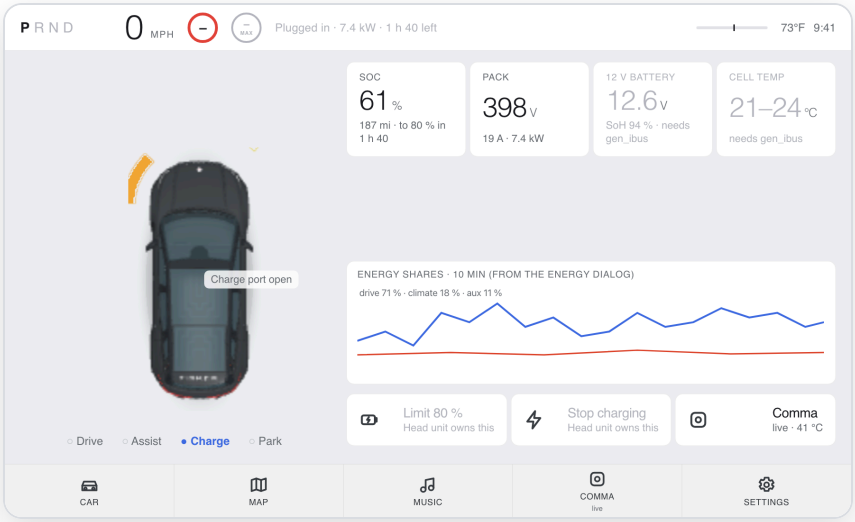
Every existing function stays one or two steps away: settings via the gear in the bar (and the tabs that are parked-only grey with the reason while driving), car controls via Car or a tap on the car, the Comma tab via the chip or the parked bar, energy via the strip's power bar, the camera window via the parked bar, replay via Playback or Comma while parked, the view menu via a long press on the scene or the Car button's popover. The bar's four quick actions were chosen from what the catalog already sends and what the stock button bar below the screen does not already cover; the stock bar has physical temperature, fan, front and rear defrost and volume keys, so duplicating fan and temperature on screen would be redundant if those keys still reach the ICC under Pulse, which is unverified (Fisker manual, Central Touchscreen Overview). Against the numbers: the essential set sits in one 96 px band (about 19 mm) at the top; bar targets of 160 × 112 frame px are roughly 32 × 22 mm, triple the minimum; climate and lock are one step; the band's single-alert rule removes the three-stacked-alerts case; the strip and bar are static DOM so the frame budget is untouched; and the collapse of music and map while driving is NHTSA's "dimming, blanking or repositioning" of non-safety dynamic information applied literally (NHTSA Guidelines, sec. V.L). The trade-off is that the world view loses about 11 % of its height to the strip and bar, and that a driver who wants album art or the big map while moving cannot have them.

Option B: Tile Dash (S3XY-style widget dashboard)

The Tile Dash gives the 3D view the top 40 % of the portrait screen under the same strip and fills the rest with a three-column grid of stat and action tiles on pages that switch by state, as the S3XY Dash does on Drive, Autopilot and charging (enhauto tutorial). The Drive page shows range, power/regen, lead distance and time, state of charge, outside temperature and trip; the Assist page (comma engaged, or Fisker ACC active) swaps in set speed, gap, MADS state, lane confidence and the speed-limit-ahead from mapd; the Charge page (parked and plugged) shows kW, time left, port state, pack current and voltage, 12 V (once received) and the energy shares that today live in the Energy dialog; the Park page shows doors, windows, lock, tyre lamps and the comma. Action tiles (seat heat, defrost, lock, heated wheel, fold mirrors) show their state on the tile and send through the same catalog entries.



Drive page: the 3D world in the top 40 %, three stat rows and one action row, page selector



Charge page in landscape: pack figures, the Energy dialog's shares, owned actions greyed with the reason

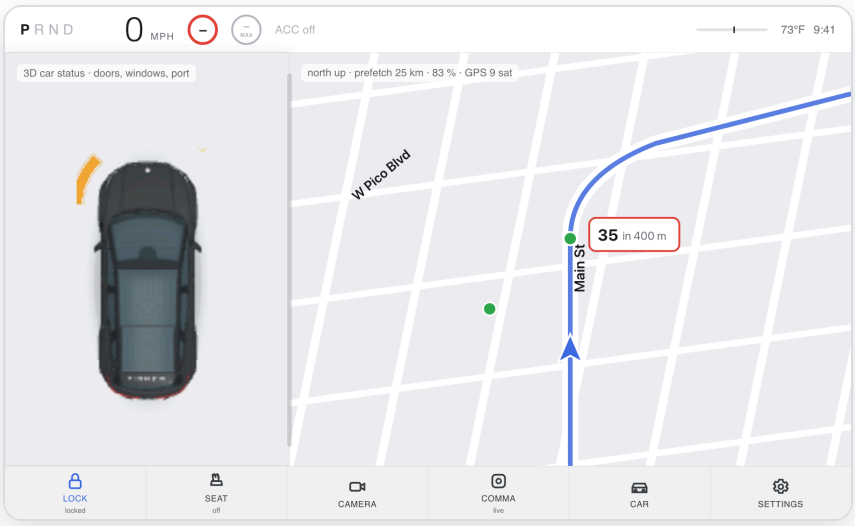
Functions are reached by page rather than by sheet: the Energy dialog becomes the Charge page, car controls stay behind Car or a tap on the small scene, settings behind the gear, the comma behind its tile. The guideline trade-off is double-edged. Tiles make every target large (360 × 300 frame px, about 70 × 60 mm) and every action one step, and a smaller canvas lightens the GPU on the head unit whose 815k-triangle model is a known worry (`CAR_CONTROLS_PLAN.md#L137–L144`). But nine numbers on a driving page invite nine glances, and the research warns that more simultaneously active elements measurably raise load; NHTSA's 12 s task budget is about tasks, not about reading a dashboard, so the risk here is the long distraction (a 3-4 s stare) rather than a failed task test (Ergosign STADT:up). The right use of this option is the parked and charging pages and the Assist page, with the Drive page capped at three tiles; a user-arranged grid with persistence in `localStorage` already has a settings store to live in (`static/js/main.js#L18–L31`).

Option C: Map Centre (map-centred layout)

The Map Centre inverts the current hierarchy: the worker's 2D street map, today a 320 px non-interactive card redrawn at most every 100 ms, fills the middle of the screen heading-up, and the 3D world becomes a Tesla-style "car status" window at the bottom whose height the driver drags between one third and nothing (`static/js/minimap.js#L25–L37`; Tesla Model Y manual). The next-turn band and road name sit between the strip and the map, the matched road ahead is drawn in accent, traffic signals as dots, and the map's own speed limit and mapd's speed-limit-ahead with its distance appear on the map as secondary, differently styled values. Parked, the map becomes the full-screen search-free overview with the car-status window hidden; in landscape the map takes the right two thirds and the 3D car the left third, which is the Tesla and Rivian split.



Driving: the 2D map fills the middle, the 3D world is a draggable car-status window at the bottom third

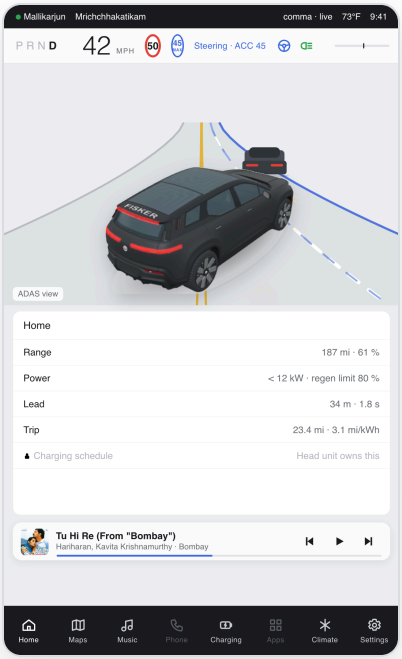


Landscape, parked: the Tesla / Rivian split, 3D car status left, map right

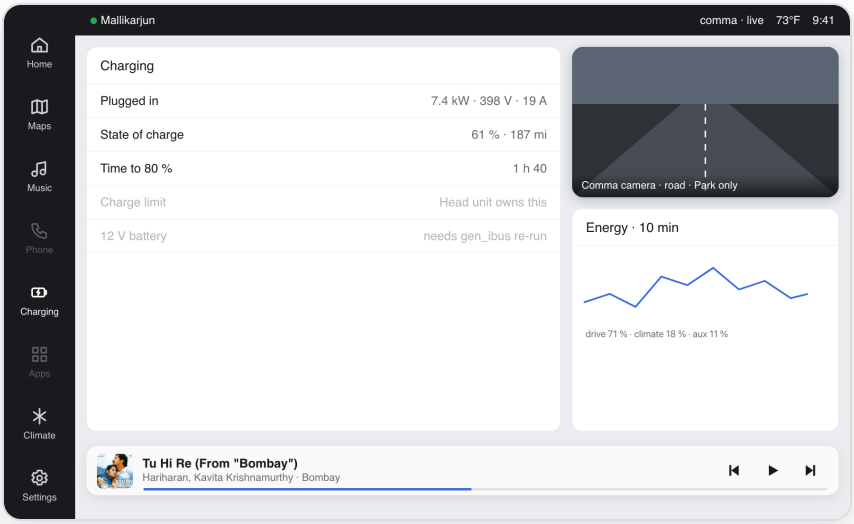
This option is the one most aligned with NHTSA's text (maps are exempt from the image lock-out; 3D detail is "not recommended") and with Mercedes' always-visible map, and it works from the car alone because the map needs only the TBOX fix and cached tiles (NHTSA Guidelines, sec. V.F.3.c; `INTERNALS.md#L236–L241`). Its costs are specific. The mini-map was throttled to 100 ms redraws to protect the frame budget, and a 1080 px wide canvas redrawn ten times a second alongside the 3D view is a cost nobody has measured on the head unit; the map is only as good as the prefetch (default 25 km radius, every 3 km or 2 min) and shows "GPS lost" when the fix is stale; and the navigation band depends on the Android notification listener, whose source app on the car's head unit is unknown and untested (`static/js/minimap.js#L1–L9`; `INTERNALS.md#L302–L308`; `android/README.md#L97–L100`). Most of all it demotes the feature that makes this app different from Pulse's own launcher, which sells CarPlay, streaming and cloud telemetry and claims no driving view at all (EIN Presswire). That argues for offering it as a mode (the "3D" bar button toggles A and C) rather than the default.

Option D: Ocean Native (stock-mimicking layout)

The Ocean Native layout reproduces the stock Control Mode skeleton so the page feels like the car: a status bar at the top (profile, app name, comma and bridge state, clock), the “ADAS and camera views” region holding the 3D world with the essential strip overlaid on its top edge, an active window beneath it (Home tiles, the car-controls sheet, Energy, Settings), a media/phone row, and the stock main-menu order along the bottom: Home, Maps, Music, Phone, Charging, Apps, Climate, Settings (Fisker manual, Central Touchscreen Overview). Items the page cannot provide (Phone, Apps) stay in the menu greyed with the reason that they are Pulse’s own launcher’s job, which is the catalog’s rule applied to navigation rather than to controls.



Driving: stock status bar, ADAS view with the strip overlaid, active window (Home), media row, stock menu order



Landscape, parked (stock Hollywood: menu on the left): Charging as the active window, comma camera, energy

Reach is the stock reach: Climate and Charging are one tap, car controls are Home then a category (two steps), settings one tap. The eight-item bottom menu gives 135 frame px per item (about 27 mm), fine against the 10 mm rule but tighter than Option A’s six. The trade-offs are about trust rather than numbers. Owners’ complaints about the stock UI are overwhelmingly about reboots, screens going dark and crashes on rotation, not about layout, and reviewers call the layout “relatively intuitive”, so copying the structure gains familiarity without inheriting the bugs (SlashGear; Auto Express). But a page that looks like FiskerOS invites the expectation that Phone, Apps and Charging schedules work, and the catalog already has fourteen MOCK controls and ten NO_MSG ones that exist only to mirror the manual; a native-looking layout multiplies the greyed rows the driver has to read past (`static/js/carcatalog.js#L49–L52`). The honest use of this option is as a skin: the status-bar contents, the menu order and the day/night tokens, laid over Option A’s structure.

The options against the guideline numbers

Criterion (source)	A Glance Strip	B Tile Dash	C Map Centre	D Ocean Native
Essential set in one band near the cone (Euro NCAP 30°)	96 px strip, top	same strip	same strip	strip overlaid on the ADAS region, under a status bar

Criterion (source)	A Glance Strip	B Tile Dash	C Map Centre	D Ocean Native
Glance load while driving	lowest: one alert, one line of music, map pill	highest: 6-9 numbers plus scene	map plus small scene; nav band	scene plus Home tiles plus media row
Touch targets (10 × 10 mm, 4 mm gap)	bar 32 × 22 mm; fixes the 44 px music buttons	tiles 70 × 60 mm	bar as A; map non-interactive	menu items 27 mm wide
Steps to climate / lock / settings (max 2)	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1	1 / 2 / 1
Lock-outs while not in Park	camera, comma, replay off the bar; tabs greyed	Charge and Park pages hidden by state (not greyed)	map stays; 3D window may drag to 0	Phone/Apps greyed always; camera Park-only
Works without comma / bridge	yes; comma items absent, bridge items greyed	yes; tiles show "–" with reason	yes; needs tiles and TBOX fix	yes; more greyed rows
New CAN sends	none (0x4E, 0x528, 0x530)	none	none	none
Frame budget on the head unit	neutral (DOM only)	lighter (smaller canvas)	heavier (second large canvas)	neutral
Risk	world view loses 11 % height	stare at numbers; pages hide rather than grey	demotes the differentiator; map cost unmeasured	mimicry raises expectations

The Tile Dash's page switching hides content by state, which is at odds with the project's grey-not-hide rule; if it is used, the page selector should keep every page tappable and show why one is not current. Option C's draggable divider is the one gesture that competes with OrbitControls on the scene, and the tap-versus-drag thresholds (8 px, 300 ms) would need an exclusion zone along the divider (`static/js/main.js#L20`).

Received-but-unshown signals fund most enhancements without a bus change

The app receives 114 of the 265 messages in the IBUS DBC and decodes about 703 signals from them, but only about 167 signal names are referenced by any main-thread module; the worker's semantic snapshot has three whole groups (`dms`, `camera`, ADASIS `horizon`) with no reader at all (`research_notes/Pulse%20HUD%20UX%20options/available_data.md`; `static/js/world/fisker_world.js#L546–L581`). The bridge's 20 Hz ticks and the map tiles carry more that is extracted and dropped. The table groups the enhancements by what they cost, names the slot each takes in Option A, and flags what the sources mark as unverified or wrong.

Effort	Enhancement	Signal or extract (source)	Where it lands (Option A)	Flags
Page only	Remaining range and validity; instant and accumulated consumption; power and regen limit % (cold pack)	`VCU_DrvgMilg`, `VCU_DrvRngVld`, `VCU_InsntEgyCnse`, `VCU_DrvPwrLimPerc`, `VCU_EgyFbPwrLimPerc` in `VCU_0x503–0x505` IBUS2 (`research_notes/Pulse%20HUD%20UX%20options/available_data.md`)	range in the strip's parked state and the Charge page; limit marks on the power bar	replaces the Energy sheet's mock "330 mi × SoC" range (`static/js/carcontrols.js#L51`); owners distrust the stock range figure (oceanforums)

Effort	Enhancement	Signal or extract (source)	Where it lands (Option A)	Flags
Page only	Trip distance and average consumption; odometer	`VCU_0x5EA` IBUS1 500 ms; `BCM_TotMilg_ODO` in `BCM_0x641`	Drive page tile; Service rows (replacing mock VIN rows)	none
Page only	Charging fault reasons and “equipment power not enough”	`VCU_0x363` IBUS2 (31 flags, all unused)	Energy status line when charging fails to start	matrix-only; never verified on a car
Page only	Driver attention from the car’s own camera: drowsiness level, attention zone, camera blocked, DCAA warning	`0x527` into `dms` (`static/js/world/fisker_world.js#L546–L559`)	eye telltale in the strip (today only the comma’s `distracted` feeds it); “driver camera blocked” notice within 10 s, repeating every 10 min per Euro NCAP	works without the comma; enum meanings from the matrix
Page only	Red-light warning and green reminder; blinking state	`tlr.warning` (`ADAS_TLR_WarnReq`), `tlr.status`	band alert; the fork users’ green-light chime, from the car alone	none
Page only	Camera blinded or faulted; ambient light class	`camera` group (`ADAS_FrntCamBli`, `ADAS_FrntCamFlt`, `ADAS_AmbLi`)	degraded-sensor notice; `AmbLi` as a second input to Auto theme	none
Page only	ISA/TSR extras: supplementary plate, confidence, condition, overspeed, limit end distance	`tsr.addOn`, `confidence`, `condition`, `overspeed`, `road.speedLimitEndDist`	limit sign grows on overspeed (Tesla’s pattern), plate under it, dimmed when low confidence	none
Page only	Door-open warning (DOW), rear AEB, curb warnings per wheel, APA availability	`threats.dowState`, `aeb.rearAeb`, `parking.curbWarning`, `parking.apa.available`	parked-state band and PDC arcs	APA flow itself stays a mockup
Page only	GPS quality: satellites, HDOP, fix OK	`TBOX_0x46F/0x472/0x473` IBUS1	explains the map’s “GPS lost” chip	IBUS1 only, so never via the comma
Page only	Road grade; hazard lights; wiper speed; rear belts and occupancy	`ESP_RoadSlop(+Vld)`, `BCM_DangerAlrmLampSwtSts`, `BCM_FrntWiprSpd`, `BCM_0x364`	ground tilt; hazard flash on the model; per-seat belt lamp while driving	none
Page only	Lead brake light; lane width; oncoming flag per lane	`objects[].brake`, `road.laneWidth`, `road.oncoming`	red lamps on the lead car; oncoming lane shading	none
Page only	ADASIS horizon as fallback road name, speed-limit class, lane index, tunnel/bridge, slope	`horizon` from `ICC_0x255` et al. (`static/js/world/adasis.js#L129–L146`)	road name when tiles are missing; cross-check for the OSM match	header bit offset corrected by hand; curvature is a fitted conversion from two drives; sign-type table missing (`tools/gen_ibus.py#L38–L42`; `INTERNALS.md#L70–L72`)
Page only (tick)	Map speed limit ahead and distance; limit source and assist state; stop and lead flags; lane-depart flags; DM awareness, face, lockout	`liveMapDataSP.speedLimitAhead(+Distance)`, `longitudinalPlanSP.speedLimit.*`, `longitudinalPlan.shouldStop/hasLead`, `carControl.hud.*`, `driverMonitoringState.*` (`openpilot/openpilot/sunnypilot/webhud/extract.py`)	“35 in 400 m” under the limit sign; lead chevron metrics	comma only

Effort	Enhancement	Signal or extract (source)	Where it lands (Option A)	Flags
Page only (map)	Map speed limit as a secondary value; advisory speed and hazard text as curve warnings; turn-lane arrows	`way.maxSpeed`, `advisorySpeed`, `hazard` (`static/js/world/osmtile.js#L131–L154`); `turn:lanes` (`static/js/roaddetail.js#L92–L114`)	secondary limit on the map; band warning	authors chose the camera sign over stale map limits (`static/js/world/state.js#L382`); arrow drawing unverified
Page only (Android)	Device fix accuracy, provider and age; media app name and actions; "running without root" indicator	`webhud:location` fields (`android/app/src/main/java/ai/sunnypilot/webhud/DeviceLocation.kt#L73–L84`); `HudListener.kt` keys	Position status line; music line; About tab	none
Page only (catalog)	Driver and passenger set temperature	`ICC_DrvrTSet` / `ICC_PassTSet` in allow-listed `ICC_0x530`; read-back in `ECC_0x373`	Climate grid row 3, replacing the demo	the catalog says "no message for them"; the seat-move precedent (frame went out clean, seat did not move) means this needs a car test (`static/js/carcatalog.js#L588–L592`)
gen_ibus re-run	12 V battery SoC, SoH, voltage; DC-DC	`VCU_IBS_BZE1` 0x2AC IBUS1; `DCDC_0x520`	strip (parked) and Charge page	inside driver-default filter ranges; Pulse sells 12 V monitoring and the FOA ties handle failures to a low 12 V (fiskeroa buying guide)
gen_ibus re-run	Tyre-pressure warning lamps per wheel	`ESP_0x390`	Service tyre chips	lamps only: **no pressure or temperature exists anywhere in the IBUS DBC**, so the "42 psi" chips can never be real
gen_ibus re-run	Heated-wheel and auto-hold read-back; EPB; hill hold	`BCM_HeatedSteerWhlSt` in `BCM_0x386`; `ESP_0x332`; `ESP_0x268`	the heated-wheel control gains state; Auto hold moves from NO_MSG to Reads only	none
gen_ibus re-run	Inside temperature and particle concentration; cell temperatures	`ECC_0x582`; `BMS_0x382`	climate card; Charge page	none
gen_ibus re-run	Steering-wheel and stalk buttons	`MFS_0x514` IBUS1 50 ms; `CIM_0x310` IBUS2	a physical way to switch A/C modes or pages without touch (Euro NCAP "direct physical input")	the sunnypilot port uses `MFS_RiBtnSouth` as the MADS toggle, so a mapping must not fight it (jos-technion port)
gen_ibus re-run	Screen rotator; display speed, unit and odometer on IBUS; units and temperature unit; clock	`Rotator_0x587`; `ICC_0x531`; `ICC_0x529`; `TBOX_0x62F`	orientation detection; regional settings follow the car without the comma	`ICC_0x531` reaches the page today only when the comma streams it (`tools/gen_world_dbc.py#L29`)
gen_ibus re-run	Head unit's own turn text and guidance	`ICC_0x612`, `ICC_0x344` (`ICC_NavMsgText1`)	nav band without the notification listener	stock connected nav is dead post-bankruptcy, so likely empty (InsideEVs FOA)

Effort	Enhancement	Signal or extract (source)	Where it lands (Option A)	Flags
gen_ibus re-run	Trailer and hitch; phone-key distance and battery	`TRM_0x348`; `PKC_*`	parked band	`BMS_0x24A`, `iBooster_0x119`, `VCU_0x1A0` fall outside the driver's default IBUS2 ranges and would widen the shared hardware filters (root; keep `DRIVER_DEFAULTS`)
Bridge change	Turn and curve speed states, final speed limit, green-light and lead-departure alerts, DM pose and distraction type, device temperatures and battery	`smartCruiseControl`, `speedLimitFinal`, `e2eAlerts`, `distractedTypes`, `DeviceState` fields not in `extract.py`	Assist page; band chimes	one-file change on the comma, but the webhud commit stays local and is never pushed
Bridge change	Driver steering torque, rear wheel speeds, demand power without the comma	`EPS_0x1C4`, `ESP_0x116`, `VCU_0x102` are ADAS-bus only	power trails from the car alone	needs the ADAS bus, i.e. the comma or a tap
Hardware	Radar tracks with velocity	private radar-ADAS SecOC link; draft `fisker_ocean_mrr.dbc`	fused objects	the port sets `radarUnavailable = True`; DBC scales for `DistLat`, `ArelLong`, `*_Std` and `Quality` are open, no elevation so overpasses read as stopped cars (`INTERNALS.md#L134–L137`)
Hardware	Full ADAS bus and clean frames without the comma	moistpope's ICC gateway board; Pulse's SPI/CPU0 frame loss	everything above at full rate	Pulse drops half of IBUS1 and two thirds of IBUS2 (`android/README.md#L151–L154`)

Two caveats cut across the table. Whether the gateway actually broadcasts each "not received" message on the stated bus at the stated cycle is the matrix's claim, not a measurement; a bus capture should precede the `gen_ibus.py` re-run, which also needs the FM29 matrices that are not in the repo (`CLAUDE.md`). And the ADAS geometry the HUD draws rests on corrections the authors still call provisional: object distances decoded at 0.2 m/bit against the matrix's 0.5, a range correction of x / 0.80 – 3.7 with a 1.35× lateral factor "until a tape-measured check settles them", and lane-curvature and object-heading signs exposed as toggles because the matrix does not document them (`tools/gen_world_dbc.py#L37–L47`; `INTERNALS.md#L115–L127`; `INTERNALS.md#L351–L353`). The one enhancement the research argues against is drawing the lane-change trajectory `assist.lcaTrajectory` or any other agent's predicted path: prediction visualisations were the only condition rated worse and more loading in the Ulm study (Colley et al. 2022).

Build the lock-out and the strip first, then borrow pages from the other options

The recommended order follows cost against guideline weight. First, the driving policy: a `driving` flag in the worker snapshot from `VCU_GearSig` with the speed and offline fallbacks, applied on the page to close the camera

window, grey the four comma-dependent tabs and the Debug command box with “Parked only”, block the auto-tailing log and the demo drive, and give every `rows` group the same lock; this is page-only, needs no new signal, closes the largest gap against NHTSA’s per se list, and is the first thing a reviewer of the app would ask about. Second, the strip and bar of Option A: build DESIGN.md’s unbuilt 96 px status strip from the existing status-card fields, add the 112 px bottom bar with the four allow-listed quick actions plus Car and the gear, collapse music and the map while driving, enlarge the music transport buttons to at least 72 frame px, and convert the three-stacked alerts to the single openpilot-ladder band. Third, the page-only enhancements that fit the strip without adding glances: range and its validity, the car’s own driver-monitoring state and camera-blinded notice, the red-light warning and green reminder, the speed-limit-ahead and overspeed growth of the sign, and GPS quality under the map chip. Fourth, the parked surfaces: Option B’s Charge and Park pages (with the Energy dialog’s content and the 12 V figure once received) and Option C’s full map as parked pages and as the landscape variant, both reachable from the parked bar. Fifth, the `gen_ibus.py` re-run after a bus capture, prioritised by 12 V, heated-wheel and auto-hold read-back, TPMS lamps, the rotator and the steering-wheel buttons. The Ocean Native option contributes its status-bar contents and the day/night tokens, nothing structural, and the radar tap stays where the port’s author left it, as hardware work beyond the driving stack’s own needs.

The questions only the car can answer determine the numbers above and should be settled in the first session on Pulse.

Question	Why it matters	How to settle it
Orientation and resolution of Pulse’s Android surface in Control Mode	decides portrait-first and every mm figure (10 mm ≈ 51 px assumes 1080 × 1920 on 17.1 in)	`adb shell wm size`, `wm density`, `dumpsys display` with the panel in Control and Hollywood
Does the panel and the WebView rotate under Pulse, and does the Park interlock still apply	decides whether landscape variants are ever shown while moving	rotate via the Home/Rotate key with Pulse active; try to shift out of P
Is `Rotator_0x587` broadcast on IBUS	orientation detection without Android	`candump` on `can1`/`can2` for 0x587
Viewing angle from the driver’s eye point to the top and bottom of the panel	which band is inside Euro NCAP’s 30-degree cone	inclinometer from the ocular reference point, seat in a typical position
Do the physical temperature, fan, defrost and volume keys still reach the ICC while Pulse owns the picture	decides whether the bar must duplicate climate	press them with the HUD on screen and watch `ECC_0x373`
Does the BCM refuse window, sunroof and liftgate requests in motion	decides whether the climate and window cells also lock	send `ICC_*WinCtrl` at walking pace in a safe place and watch `BCM_0x343`
Does `ICC_0x530` with `ICC_DrvrTSet` set the temperature, or is it like the seat move	decides whether the demo temperature row becomes real	one request, watch `ECC_0x373`
Which messages in the “not received” list are actually on the bus and at what rate	scopes the `gen_ibus.py` re-run and the hardware-filter change	60 s `candump` per bus with the filters widened under root
Cost of a full-width 2D map canvas and of the strip beside the 3D view	decides whether Option C is viable on the head unit	the fps chip’s median/worst/JS ms with the map at 1080 px
Which app emits navigation notifications on the head unit	decides whether the nav band ever fills	run Google Maps or Waze on Pulse and watch `webhud:nav`
Whether the ADAS 3.1.1 in the FOA’s Marlin bundle changes any ADAS message	the whole decoded snapshot rests on the FM29 matrices	diff a capture before and after Marlin

Conclusion

The research reframes the HUD's problem. The app does not lack information; it already decodes several times more than it draws and has a 3D engine the research says is not what builds trust. What it lacks is a policy layer (a definition of driving, a lock that greys rather than hides, a cap of one alert) and a frame around the world view (a strip in the cone and a bar at the thumb) that the guidelines, the production cars and the stock Ocean all converge on. Those two additions are DOM-only, cost no frame time, send no new CAN frames, and work with neither the comma nor the Android bridge, which makes them the rare change that improves compliance, usability and robustness at once.

The orientation question matters less than it first appears, because every source that describes the car says the driving layout is portrait and the parked layout is landscape; the design question is therefore not "which orientation" but "what does the parked page look like", and that is where the tile grid and the big map belong. The real unknowns are physical and belong to the first session on the car: the panel's resolution and angle, whether the stock button bar still works under Pulse, and whether the gateway broadcasts the messages the matrices promise. Until those are measured, the mm, cone and rate figures in this report are derived from the design frame and the matrix cycle times, not from the car.